VLSI Design Course Project

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Abstract— This project is dedicated to designing a fast and power efficient 4-bit Carry lookahead circuit. The logic is simplified by using complements of the inputs, with a 4-transistor XOR gate and a true single phase clock (TSPC) D flip-flop.

Keywords— CLA, CMOS, TSPC, NGSPICE, MAGIC, Verilog, FPGA, logic gate

I. Introduction

Addition is a basic function in ALUs. A ripple adder has a huge propagation delay as the carry of higher bits depends on the carry-out of lower significant bits. In a carry lookahead adder [1], the carry into each bit is calculated from the inputs directly. By using a modified logic that uses the complements of inputs the hardware implementation can be simplified. A 4-transistor implementation of XOR [2] has been used to reduce the power consumption.

II. Implementation of 4-bit Carry look-ahead Adder

A. Principle of 4-bit CLA

The role of CLA is to precompute carry-in for each bit using inputs. The inputs are of the form:

$A = A_3A_2A_1A_0$

$B = B_3B_2B_1B_0$

The sum output should be of the form:

$S = S_3S_2S_1S_0$

The logic for a 4-bit CLA is given by the equations:

$$C_1 = G_0 + P_0C_0$$

$$C_2 = G_1 + P_1G_0 + P_1P_0C_0$$

$$C_3 = G_2 + P_2G_1 + P_2P_1G_0 + P_2P_1P_0C_0$$

$$G = G_3 + P_3G_2 + P_3P_2G_1 + P_3P_2P_1G_0$$

$$P = P_3P_2P_1P_0$$

$$C_4 = G + PC_0$$

Where carry-propagation Pi=Ai $\oplus$ Bi and carry generate $G_i = A_i B_i$(AND logic)

B. Proposed CLA Structure

To simplify hardware implementation, we use a logic based on complements of the inputs. Let Pi' and Gi' be defined as

$$P'_i = \overline{A_i + B_i}$$

and

$$G'_i = \overline{A_i \cdot B_i}$$

In the carry equations Pi may be replaced by the OR function (Ai + Bi): XOR and OR differ only when Ai = Bi = 1, where Gi = 1 already determines the carry. XOR is kept only for the sum, Si = Ai XOR Bi XOR Ci.

$$C_1 = \overline{G'_0 \cdot (P'_0 + \overline{C_0})}$$

$$C_2 = \overline{G'_1 \cdot (P'_1 + G'_0)} + \overline{(P'_1 + P'_0} \cdot C_0)$$

$$C_3 = \overline{G'_2 \cdot (P'_2 + G'_1)} + \overline{(P'_2 + P'_1} \cdot \overline{G'_0(P'_0 + \overline{C_0})})$$

$$G'_{out} = \overline{\overline{G'_3\,(P'_3 + G'_2)} + \overline{(P'_3 + P'_2}\ \overline{G'_1\,(P'_1 + G'_0)})}$$

$$P'_{out} = \overline{\overline{P'_3 + P'_2}\ \overline{P'_1 + P'_0}}$$

$$C_4 = \overline{P'_3 + P'_2} \cdot \overline{P'_1 + P'_0} \cdot C_0 + \overline{G'_3 \cdot (P'_3 + G'_2)} + \overline{(P'_3 + P'_2} \cdot \overline{G'_1(P'_1 + G'_0)})$$

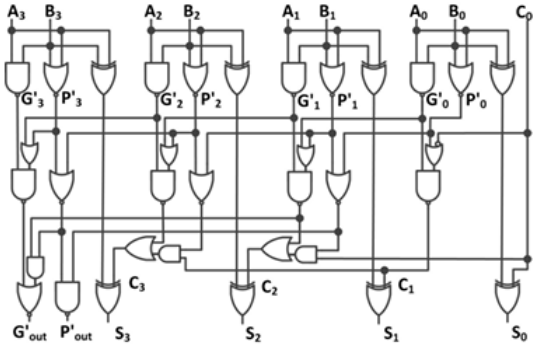
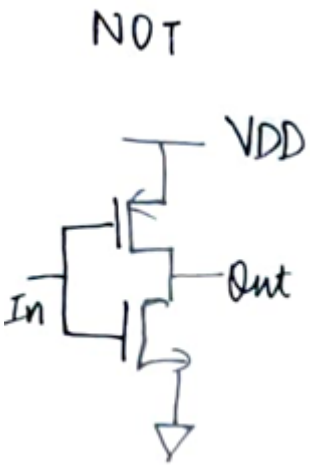


Fig. 1. Proposed circuit structure of 4-bit CLA.

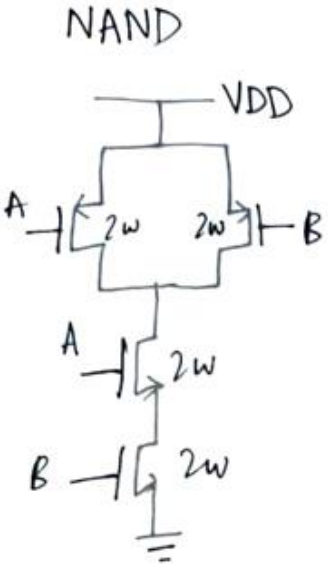
C. Design (topology and sizing)

The CLA has been implemented using Static CMOS logic and a positive edge triggered D flip flop using TSPC logic. The parameters for a minimum delay inverter are taken as L = 2λ, Wn = 6λ, Wp = 12λ = 2Wn and W = 6λ, where λ = 0.09u. We size the gates such that the worst-case capacitances of the gates match that of an ideal inverter. This minimizes the delay.

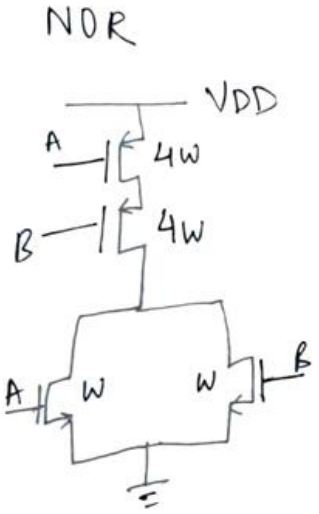
Inverter: Wn = W, Wp = 2W



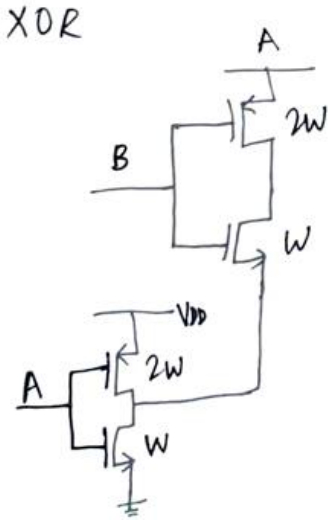
NAND: Wn = 2W, Wp = 2W



NOR: Wn = W, Wp = 4W

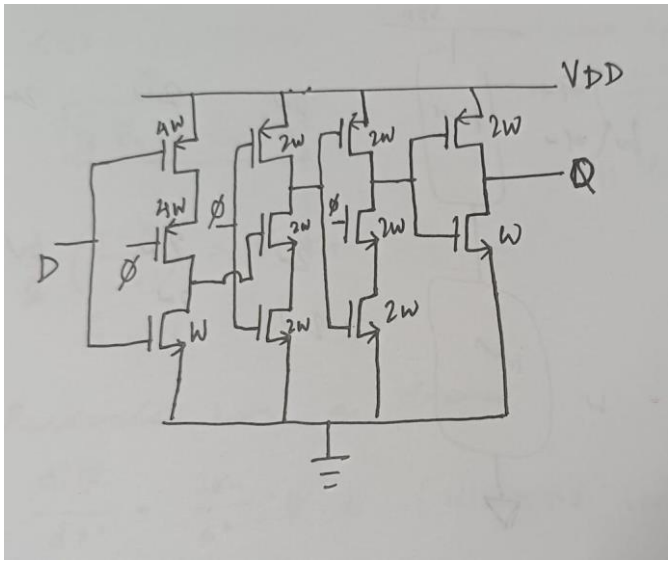


XOR: Wn = W, Wp = 2W



D Flip Flop:

Implemented using TSPC logic [3] for high speed; a single clock phase avoids skew between clock phases.



Thold= 0

Tsu ≈ 0ns

Tclk-Q= 0.06ns (rise), 0.12ns (fall)

(pre-layout, unloaded output)

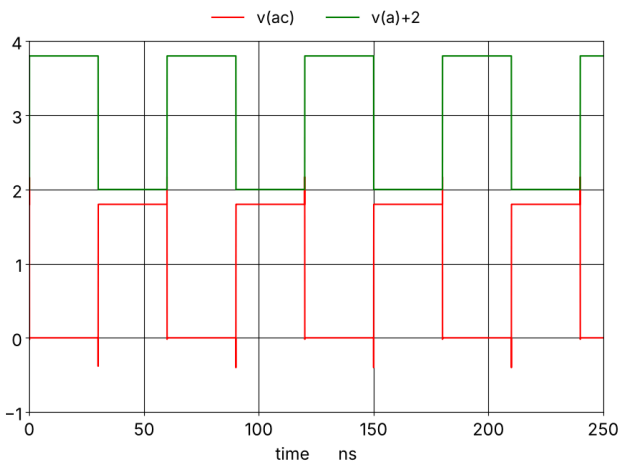
III. NGSPICE SIMULATION RESULTS

Inputs given:

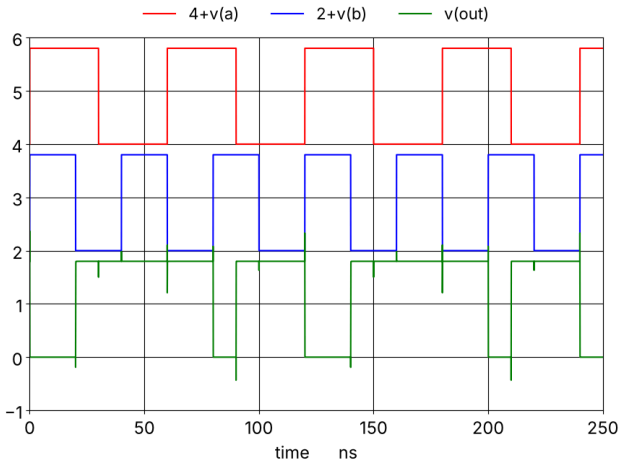
v1 A gnd pulse(0 1.8 0ns 0ns 0ns 30ns 60ns)

v2 B gnd pulse(0 1.8 0ns 0ns 0ns 20ns 40ns)

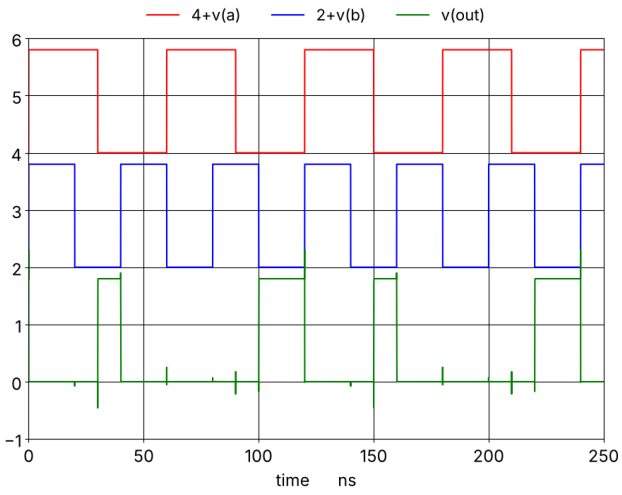
A. Inverter



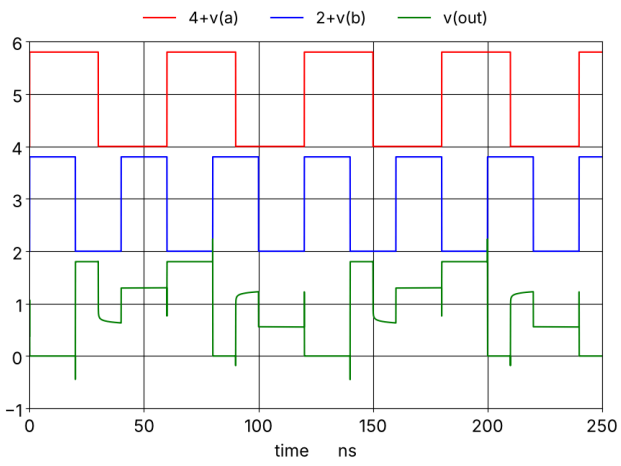
B. NAND Gate



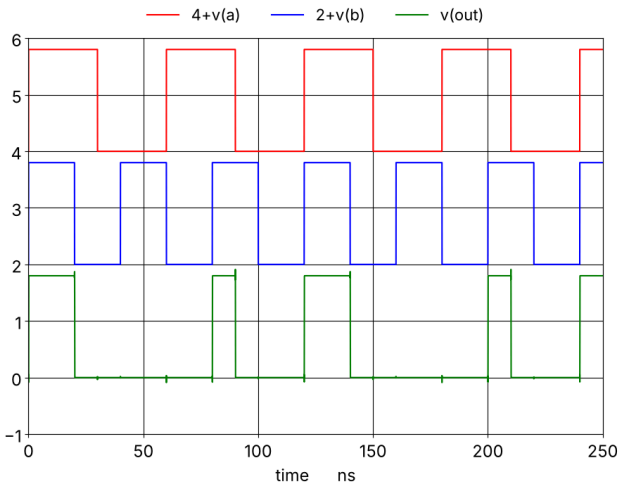
C. NOR Gate



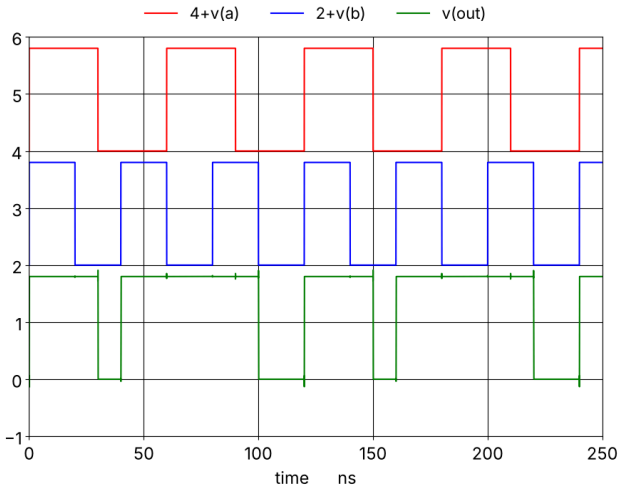
D. XOR Gate



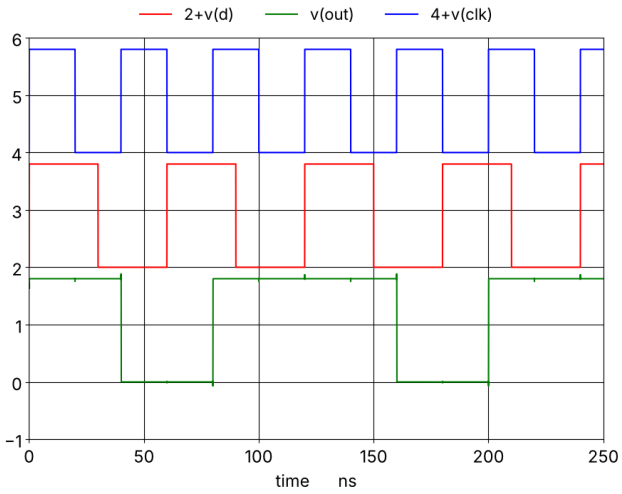
E. AND Gate



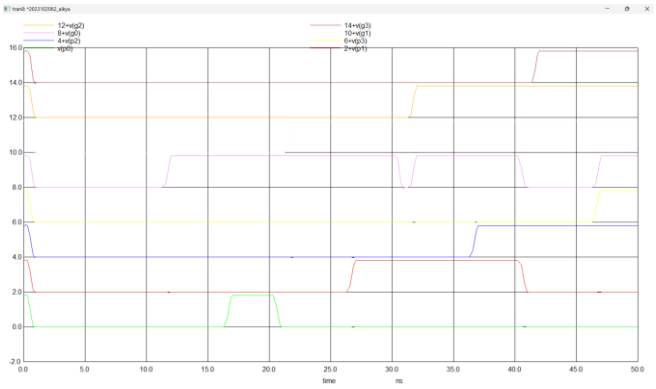
F. OR Gate



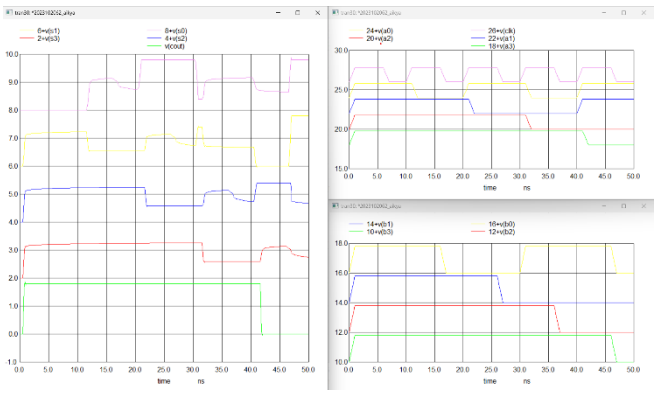
G. *D Flip Flop*



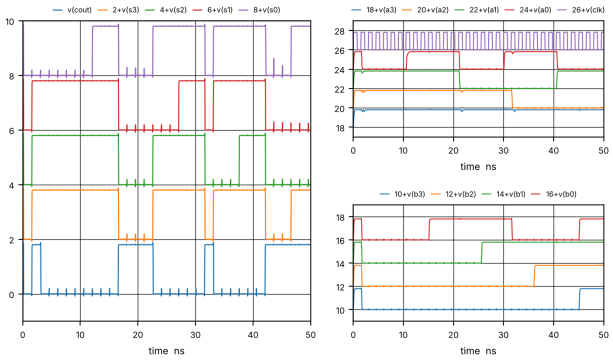
H. *Propagate Generate*



I. *CLA*



J. *Complete circuit*



Max clock frequency: 667MHz(1.5ns)

Worst case logic delay: 0.68ns

(pre-layout; correct sums down to 1.5ns clock period)

IV. STICK DIAGRAMS

Legend:

Blue: Metal

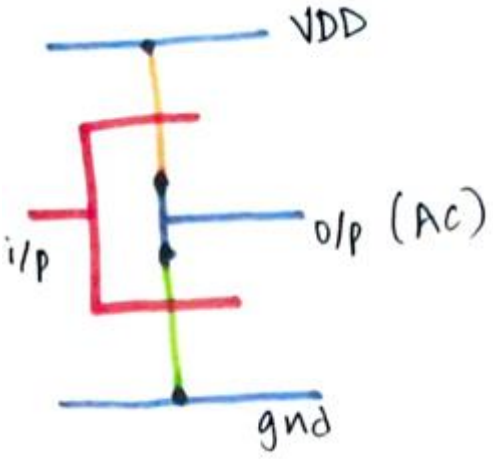
Black: Contact

Red: Polysilicon

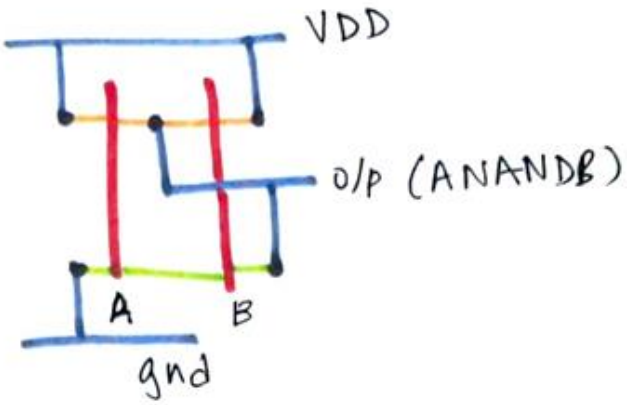
Orange: p-diffusion

Green: n-diff

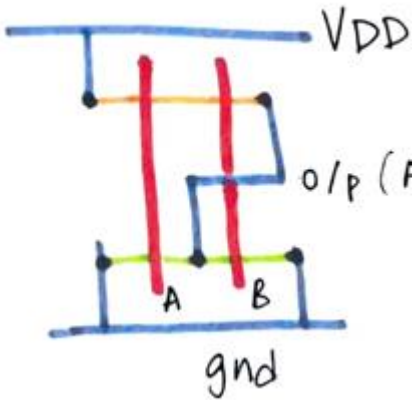
A. *Inverter*



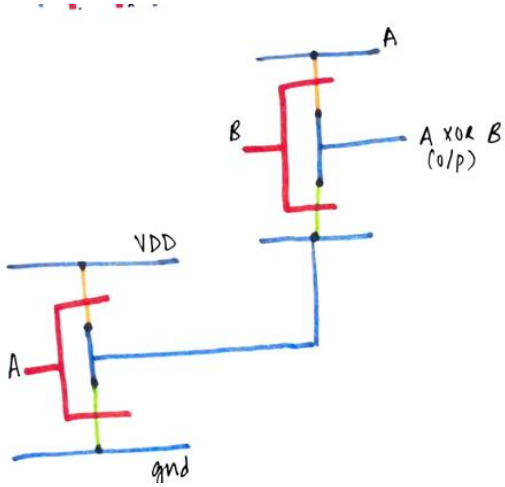
B. NAND Gate



C. NOR Gate

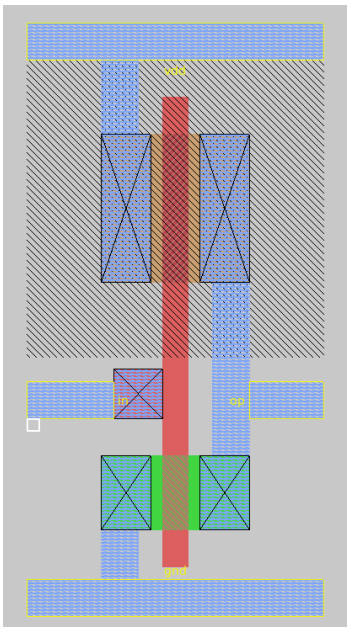


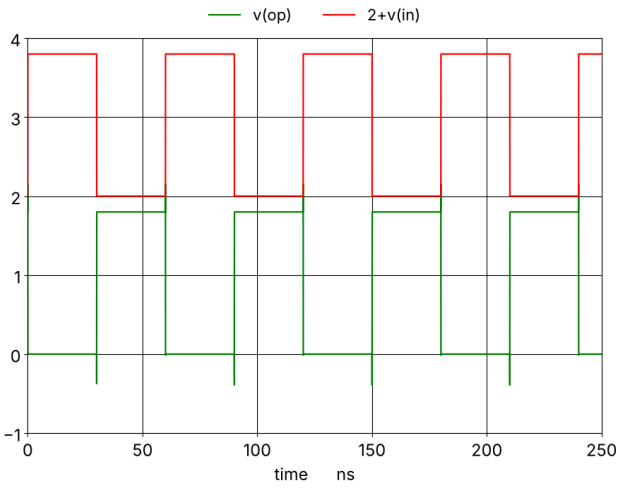
D. XOR Gate



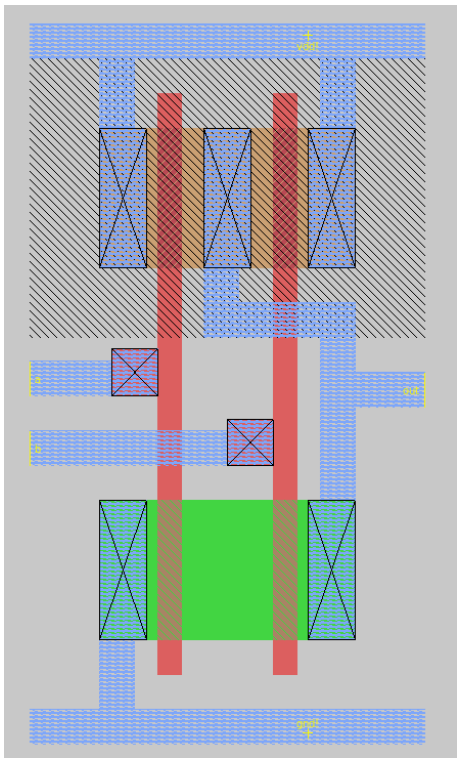
V. Post layout simulation using MAGIC

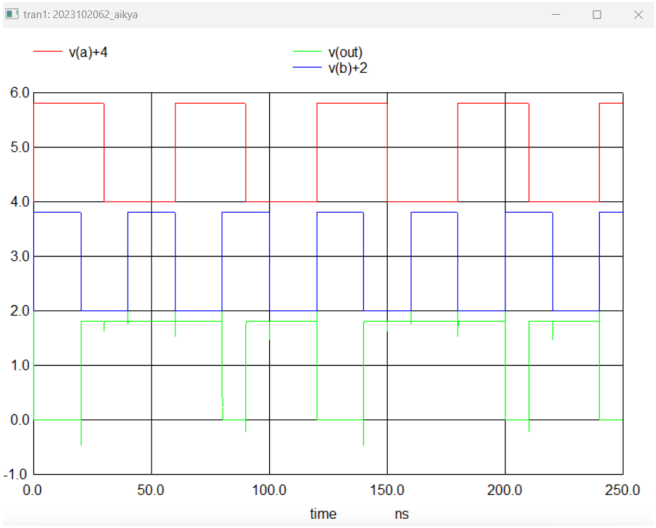
A. Inverter



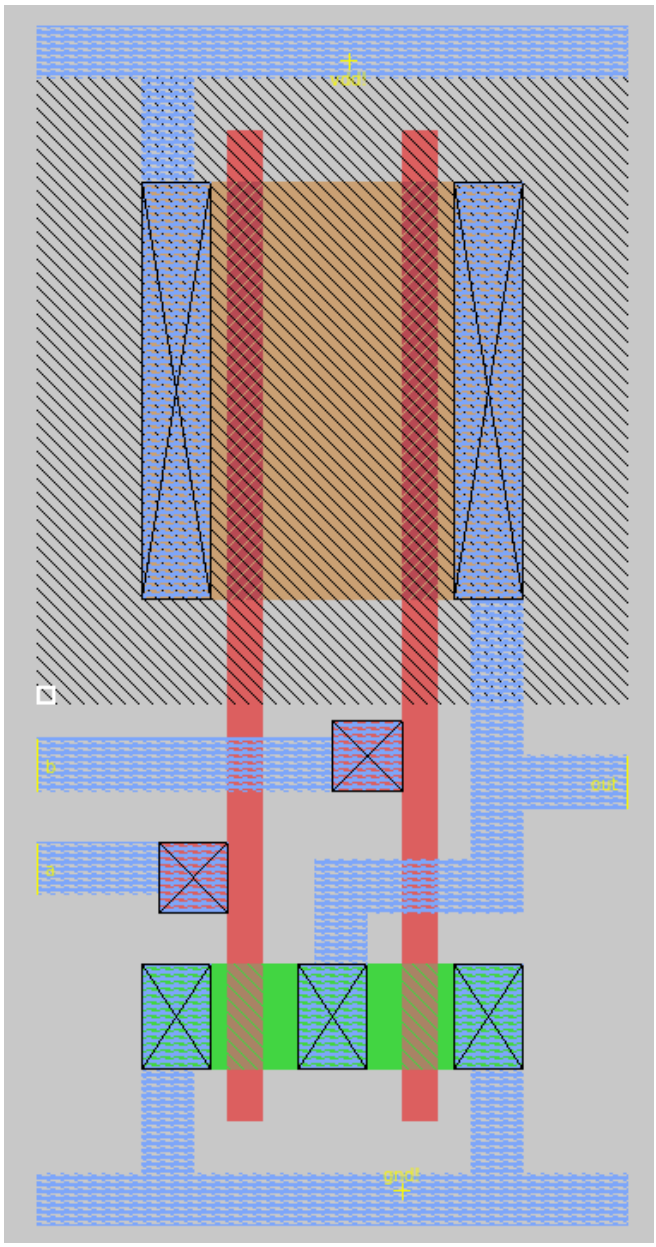


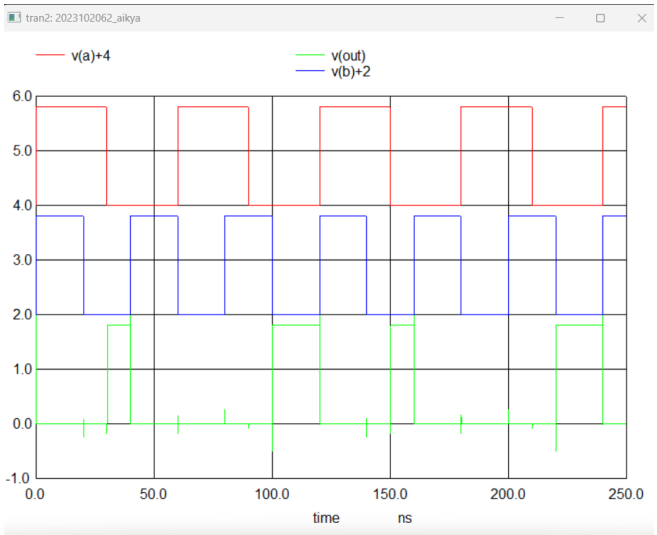
B. NAND Gate



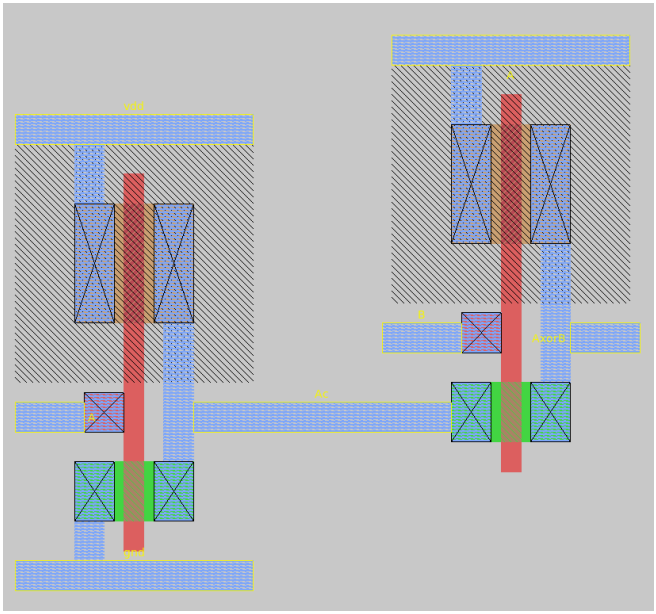


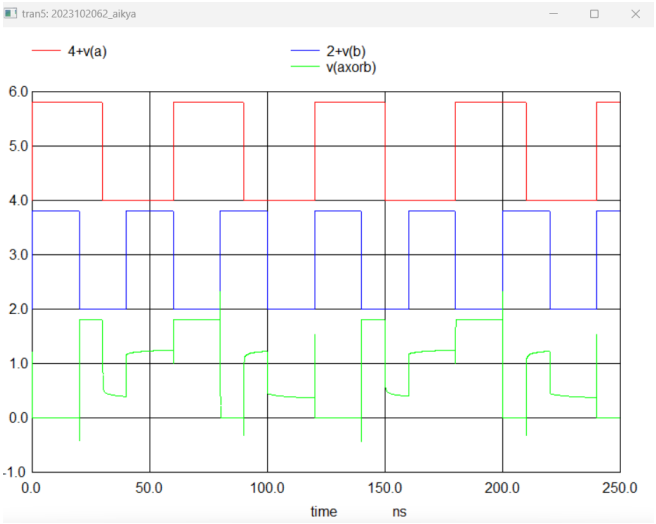
C. NOR Gate



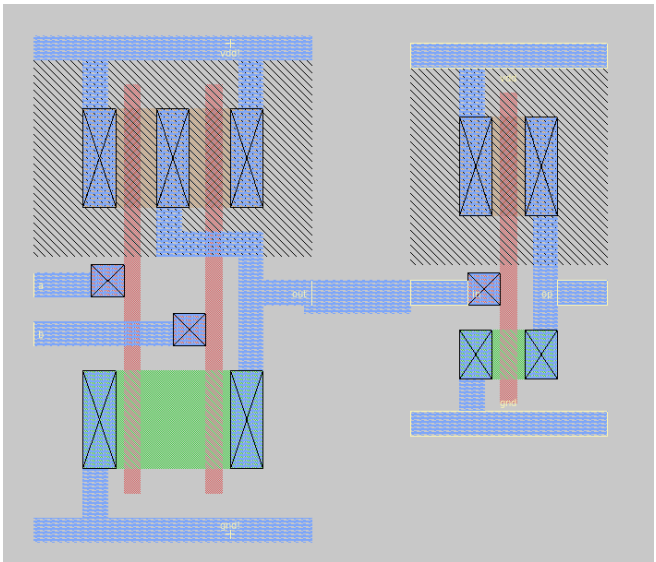


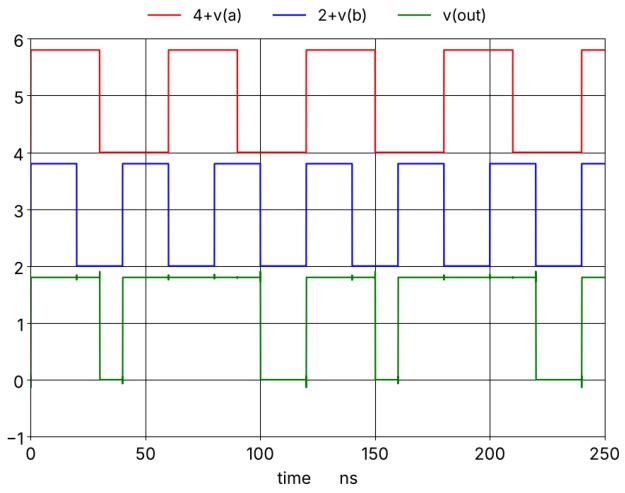
D. XOR Gate



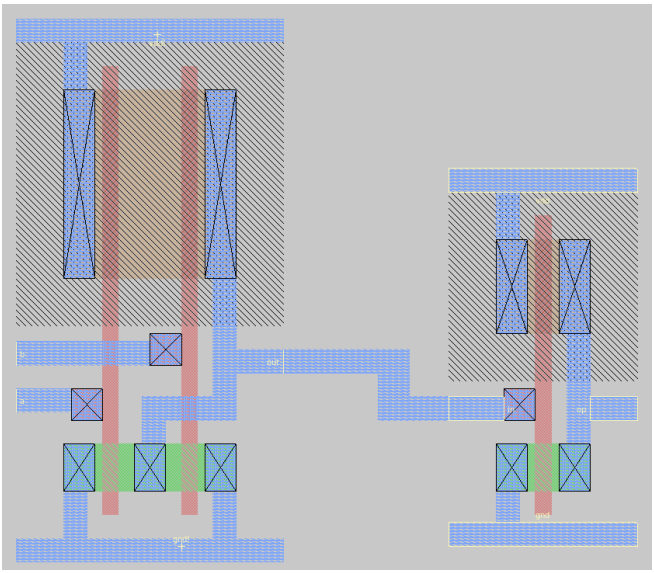


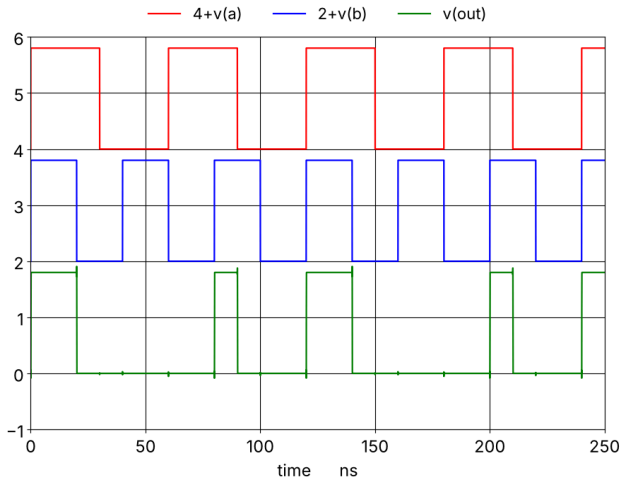
E. OR Gate



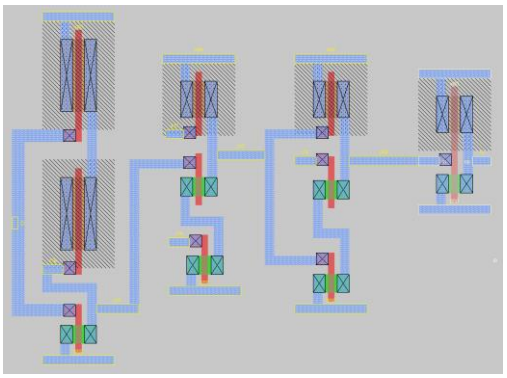


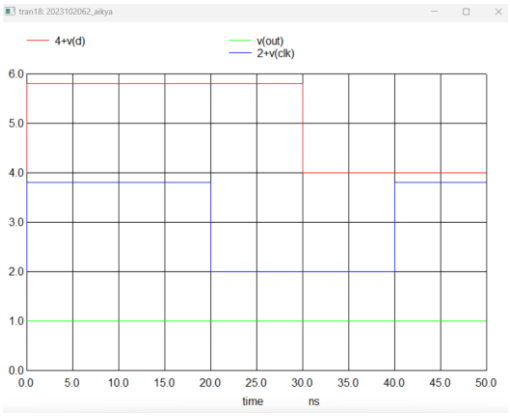
F. AND Gate



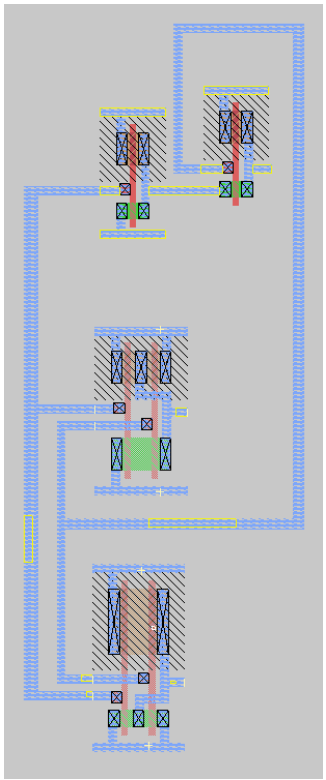


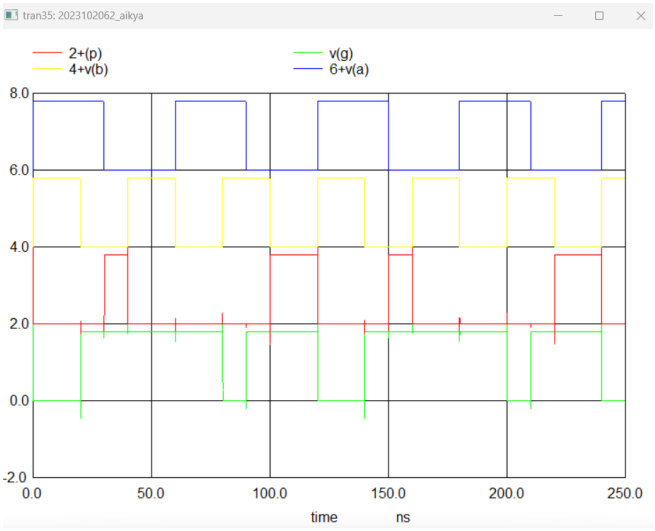
G. D Flip Flop



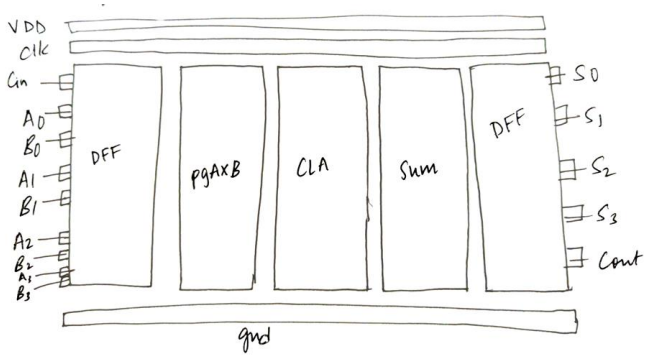


H. Propagate generate





I. Floor Plan



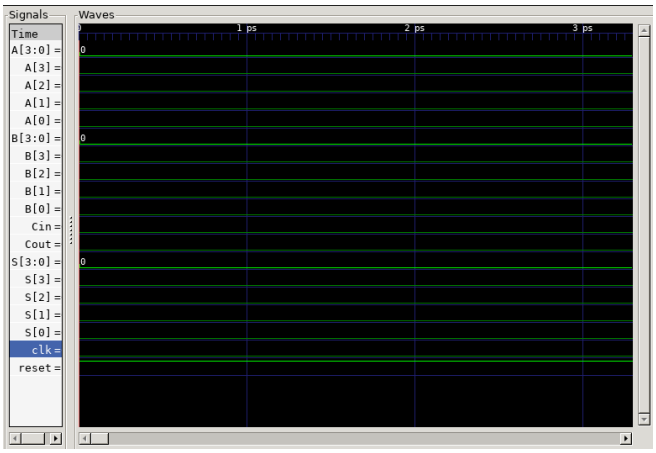
Vertical pitch: 817λ, Horizontal pitch: 1080λ

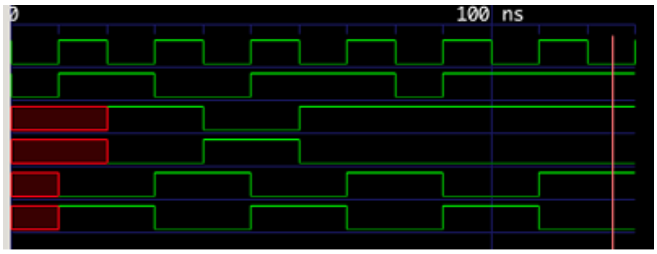
VI. Verilog

Terminal output:

```
Time: 0 | A: 0000 | B: 0000 | Cin: 0 | S: 0000 | Cout: 0
Time: 40000 | A: 0001 | B: 0011 | Cin: 0 | S: 0000 | Cout: 0
Time: 50000 | A: 0001 | B: 0011 | Cin: 0 | S: 0100 | Cout: 0
Time: 60000 | A: 0000 | B: 0000 | Cin: 0 | S: 0100 | Cout: 0
Time: 70000 | A: 0000 | B: 0000 | Cin: 0 | S: 0000 | Cout: 0
Time: 80000 | A: 1100 | B: 0011 | Cin: 0 | S: 0000 | Cout: 0
Time: 90000 | A: 1100 | B: 0011 | Cin: 0 | S: 1111 | Cout: 0
```

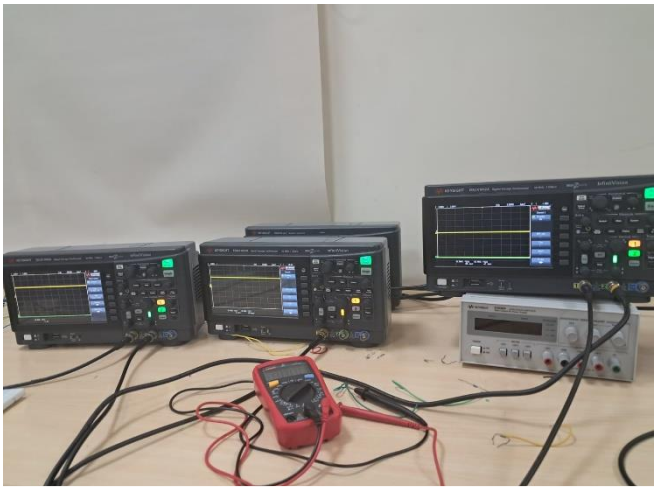
GTK Wave:





VII. FPGA

Inputs:
A=1111
B=1001
Cin=0



S0=0, S1=0, S2=0, S3=1, Cout=1

REFERENCES

[1] Jia Miao and Shuguo Li, “A Novel Implementation of 4-bit Carry Look-ahead Adder”.

[2] Anubhav Garg, Deepak Agrawal, Sachin Rajput, Smita Singhal, Anu Mehra , “A Novel Power Efficient XOR Gate based on Single Inverted Input”.

[3] JIREN YUAN AND CHRISTER SVENSSON, “High-speed CMOS Circuit Technique”.

[4] M.Morris Mano, Digital Design.